

TPC-402: The Signed Diagonal-Deletion Coefficient in a Finite C1 Panel

Liang Wang

School of Mathematics and Statistics, Huazhong University
of Science and Technology (HUST), Wuhan, China

September 5, 2026

Abstract

We continue the exact finite decomposition from TPC-401 and retain the endpoint sign as a named modeling choice. For $N < Q < p$ and an off-diagonal pair, the signed component coefficient has the exact form $M_\sigma(u, v) = T_{uv}[-A_\sigma + b_\sigma(u) + b_\sigma(v)]$. An exact rational audit checks the all-plus and alternating-index laws on 240 sampled off-diagonal rows, or 209280 prime-level comparisons, across six TPC-400 origins and the 872-prime shell. The $N = 13, Q = 8, p = 11$ anchor remains a boundary counterexample. No arithmetic sign identification or asymptotic estimate follows.

1 Setup and boundary

Use $I_o = \{o, \dots, o + N - 1\}$, $T_{uv} = H^2 / (H^2 + (u - v)^2)$, and $a_p = p(p/Q)^2 / (p - 1)$. The six origins are

7600001, 7603209, 7606417, 7609625, 7612833, 7616041.

The laws $\sigma_p = 1$ and $\sigma_p = (-1)^{\text{index}(p)}$ are synthetic finite probes. All claims here are finite.

2 Signed coefficient identity

Define

$$A_\sigma = \sum_p \sigma_p a_p, \quad b_\sigma(u) = \sum_{p|u} \sigma_p a_p.$$

For $u \neq v$ in a production window, $|u - v| < p$ and no prime in the shell divides both u and v . Thus

$$\sum_p \sigma_p a_p \mathbf{1}_{p \nmid u} \mathbf{1}_{p \nmid v} = A_\sigma - b_\sigma(u) - b_\sigma(v),$$

and the literal centered component gives

$$\boxed{M_\sigma(u, v) = T_{uv}[-A_\sigma + b_\sigma(u) + b_\sigma(v)]}.$$

The diagonal is still zero because the original operator deletes it.

3 Exact audit

The producer uses $N = 1024, Q = 8192, H = 66$, exact ‘Fraction’ arithmetic, two laws, five positions per origin, and every one of the 872 shell primes. It checks 240 sampled off-diagonal rows and 209280 prime-level equalities. The independent checker reverses the shell order, checks the typed census, and repeats a reverse-shell coefficient check; a three-mutation stress checker rejects altered contracts. All exact equalities pass.

4 Anchor obstruction and route ledger

At $N = 13, Q = 8, p = 11$, the active pair $u = 7600001, v = 7600012$ has difference -11 . The divisibility term is one, so the production identity does not apply. The finite result is therefore bounded by its hypotheses.

item	status
signed coefficient identity	PROVED_EXACT_FINITE
finite audit	NUMERICAL_OBSERVATION
source sign identification	OPEN
arithmetic advance / fixed-power credit	NO / 0
Route-B / twin-prime result	OPEN / NONE

The next clue is `TEST_C1_SIGNED_DIAGONAL_TERM_GROWING_OBSTRUCTION`.

Reproduction

The project contains the required README, code, experiments, results, notes, proof package, and `paper/paper.pdf`. The three checks are listed in the README.